A Mirror in Feigenbaum's Table

The published δ for maps of odd order, the maps' own values, and one sign that may explain the difference

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Preprint. This note has not been peer reviewed. Its search of the literature is limited, and where a source was read only in part, or not at all, the text says so. It comes from a part-time research team that works in the open and gives its tools away. Every number it computes is printed by the tool that ships with it. CC BY 4.0.

What we found, and what we ask

Feigenbaum's constant δ measures how fast a family of maps doubles its way into chaos: as the map's knob μ is turned up, each new doubling of the period needs a turn of the knob δ times smaller than the last. Its partner α measures how much each new picture of the orbit shrinks, and its minus sign says that each picture is also flipped, like an image in a mirror. For the maps $f(x) = 1 - \mu|x|^d$, whose single hump has order d, the standard table of both constants was computed by Briggs to dozens of figures (Briggs 1991; Briggs 1997, Table 2.3). Its rows for orders 3 to 6 are repeated in MathWorld's page on the Feigenbaum constant, and its order-3 value in Finch's *Mathematical Constants* (2003, Table 1.3, read in a printed copy).

McCartney (2020) computed δ for these maps directly, from their superstable orbits (the settings of μ at which the top of the hump, x = 0, lies on a cycle of period 2, 4, 8, and so on) followed to period 2^{32}. He found values that differ from the table's at every odd order, checked his own calculation four ways for numerical error, and concluded that the discrepancy more likely lay in the earlier work, without naming a cause. Repeating his reading with our own code, and reading the maps by a second route, we agree. The table's α agrees with our solution of Feigenbaum's equation at every order from 2 to 12, and its δ at every even order, but its δ at every odd order differs from what the maps do. At order 3 the table gives 5.9679687038 and the maps 6.0846890743, so the table is too low by 0.117. At order 5 the table gives 8.3494991321 and the maps 8.3443847487, so the table is too high by 0.005; at orders 7, 9 and 11 it is too high by 0.062, 0.095 and 0.116. Direct readings of the maps stand behind these values at orders 2 to 10 (McCartney's) and at orders 3, 5 and 11 (ours). At order 11 the separate helper's superstable program, run unchanged to period 2^{20}, gives δ = 12.96021250 by Wynn's extrapolation, its last two columns agreeing to 2×10^{-10}, and α = −1.2670614079. At order 12, where the table and Feigenbaum's equation agree, the same program gives δ = 13.53507567, so every row of Table 1 now has a direct reading of the maps behind it.

We think we can see how the difference arose, and we may be wrong. To get δ, Briggs's calculation, which solves Feigenbaum's equation, has to look at the hump's other side, a point that lies to the left of zero. The map uses the size of x, written |x|, so for a value to the left of zero the minus sign should be dropped before the value is raised to a power. If the power is written with its sign instead, nothing changes for even orders, because an even power of a negative number is positive anyway. For odd orders every other term flips sign, and δ comes out wrong. α comes from the fixed function alone; in our reconstruction that function is written with the size of x everywhere, the left side included, so α stays right everywhere, and only the small change to it whose growth δ measures is written with its sign. That is exactly the pattern in the table. One way a program could come to do this is by holding the fixed function as a series in $|x|^d$ while writing the change's terms as plain powers of x divided by powers of α. Briggs's thesis supports the first half and is silent on the second: it prints the collocation equation for the fixed function with the absolute value taken explicitly at the negative point (section 2.1.2), but says only that the matrix for δ is built by a similar method, without writing the change's terms. Built with that one sign, our calculation gives the table's order-3 value to 2×10^{-29}.

We have not seen the program that made the table, so this is a reconstruction that reproduces its numbers, not a reading of its code. Others noticed the odd orders before us. Vieru (2008, section 7, read through a web reader) asked whether anyone had noticed that the values for odd orders lie on a smooth curve of their own, apart from the even orders'; he knew no formula for either curve, found that one formula for both gives only wrong results, and tied the odd curve to the odd orders' maxima not being analytic. His figures plot values for the same maps without saying where the odd ones come from. McCartney (2020) found values that differ from the table's. Moon (2013), in a Wolfram Demonstration on estimating δ, notes that its values agree with Briggs's thesis at even orders but differ slightly at odd ones, and that it would be good if someone could explain this (its Details section, read as a search engine indexes it; the live page did not open here). We have not found the cause stated. We would welcome more eyes: anyone who can confirm or correct the values below, point to a correction we missed, or show where our own calculation goes wrong.

The difference turned up in *The Loop Atlas* (Schulz 2026), a book of small loop toys by the same team, deposited beside this note, when its checks of δ for the map of order 3 would not meet the table; the book carries it among its results.

Order d	δ as printed (Briggs 1997)	δ from Feigenbaum's equation, absolute powers (this note)	δ of the maps' superstable orbits (McCartney 2020)	δ with signed powers (odd d)	α (printed, and ours from the equation, agree)
2	4.6692016091	4.6692016091	4.6692016091029906718	(same as the map's)	−2.5029078751
3	**5.9679687038**	**6.0846890743**	6.084689074262121896	5.9679687038	−1.9276909638
4	7.2846862171	7.2846862171	7.28468621707334	(same as the map's)	−1.6903029714
5	**8.3494991321**	**8.3443847487**	8.3443847487310	8.3494991321	−1.5557712502
6	9.2962468328	9.2962468328	9.296246832772	(same as the map's)	−1.4677424503
7	**10.2221595288**	**10.1597798679**	10.15977986791	10.2221595288	−1.4051107883
8	10.9486242659	10.9486242659	10.9486242660	(same as the map's)	−1.3580172791
9	**11.7683336396**	**11.6731058504**	11.673105851	11.7683336396	−1.3211857598
10	12.3414090453	12.3414090453	12.34140905	(same as the map's)	−1.2915168673
11	**13.0765458057**	**12.9602125044**	(not given)	13.0765458057	−1.2670614079
12	13.5350756662	13.535075666	(not given)	(same as the map's)	−1.2465277517

Table 1. For the maps $f(x) = 1 - \mu|x|^d$. The printed values are Briggs's (thesis, Table 2.3, which prints each to dozens of figures), rounded here to 10 places; MathWorld repeats the rows for orders 3 to 6, cut off rather than rounded. Our values are read from Feigenbaum's equation, with absolute powers, at 32 and 36 coefficients and printed to the places on which the two agree, at most 10. McCartney's are his, from superstable orbits to period 2^{32}, as printed in his Table 1. The odd orders, where the printed δ and the map's differ, are in bold. The fifth column comes from the same operator built with signed powers at the point x/α, and gives the printed odd-order values. α is given with its sign; MathWorld prints its size.

How the maps' own values were read

Table 2 sets the readings side by side at orders 3 and 5. The superstable parameters are read directly from the maps: the settings of μ at which the top of the hump lies on a cycle of period 2^n, found one after another, and δ as the limit of the ratio of the gaps between them. Feigenbaum's own functional equation, $g(x) = \alpha\, g(g(x/\alpha))$ with $g(0) = 1$, is solved as a series in $|x|^d$, and δ is the largest eigenvalue in size of the doubling operator's linearisation about its solution, described in the next section. The separate helper is a fresh run of the same AI assistant (Claude), given no number and not told which result was expected, and asked to compute δ from superstable parameters and, at order 3, from the equation twice, once with each way of writing the powers; it wrote its own programs from scratch.

Reading	By	Order 3	Order 5
Superstable parameters to period 2^{14}, Aitken's step on the last three gap ratios	this note	6.08468907	8.344385
Superstable orbits to period 2^{32}	McCartney (2020)	6.08468 90742 62121 896	8.34438 47487 310
Superstable parameters to period 2^{20}, Wynn's epsilon method	the separate helper	6.08468 90742 62121 9	8.34438 47487 3
Feigenbaum's equation, absolute powers, 36 coefficients	this note	6.08468 90742 62121 89604 61210 66259	8.34438 47487 3094
Feigenbaum's equation, absolute powers, 30 coefficients	the separate helper	6.08468 90742 62121 89604 61210 66	(not computed)
Feigenbaum's equation, signed powers, 36 coefficients	this note	5.96796 87037 77451 04099 41930 19957	8.34949 91320 66964
Feigenbaum's equation, signed powers, 30 coefficients	the separate helper	5.96796 87037 77451 04099 41987 2	(not computed)
As printed	Briggs (1997), Table 2.3	5.96796 87037 77451 04099 41930 19979 672…	8.34949 91320 66963 5…

Table 2. Readings of δ at orders 3 and 5, with long decimals grouped in fives. Ours are printed by the tool that ships with this note; McCartney's are as printed in his Table 1; the helper's are its own programs' outputs, which ship beside the tool, printed as they came: its signed-power value at 30 coefficients differs from ours at 36 by 6×10^{-24}, which we take to be the limit of 30 coefficients there. Briggs's digits are copied from his page.

McCartney's values agree with ours within one unit in his last printed place at every integer order he gives (exactly at orders 3, 4, 7 and 10; one unit apart at orders 2, 5, 6, 8 and 9; our values to his full length are printed by the tool). At the non-integer orders e and π, where every program must use |x| because a negative number raised to a non-integer power is not a real number, his values and our equation's agree within 2×10^{-19} and 2×10^{-17} (his 5.713359642268103049 and 6.265069285851388; ours 5.7133596422681030492 and 6.2650692858513880150). We did not compute his other non-integer orders, √2, √3 and the golden ratio.

One earlier, rougher set of values bears on the question too. Delbourgo and Kenny (1986), as Briggs's thesis lists them (Table 2.2), gave 4.67, 6.08, 7.29, 8.35, 10.2 and 12.3 at orders 2, 3, 4, 5, 7 and 10. Their 6.08 at order 3 is the maps' value, 0.11 from the table's; their 8.35 at order 5 is what the table's 8.3495 rounds to, while the maps' 8.3444 rounds to 8.34, and their 7.29 at order 4 is off the accepted 7.2847 in its last place. Five of the six fit one reading: they are what the maps' own superstable gap ratios round to when read part way down the cascade, from the parameters of periods 32, 64 and 128 or of 64, 128 and 256, where the ratio is still 8.349 or 8.346 at order 5 and 7.286 or 7.285 at order 4 (the tool prints every depth). So their 8.35 is the maps read part way, not support for the table. Their 12.3 at order 10 needs a deeper reading; their paper, not read here, would say how they read their values.

With absolute powers, as the maps need, the values of δ lie on one smooth curve. A third difference measures how the curve's bending changes from one point to the next; on a curve that bends as gently and steadily as this one, it would be expected to keep one sign and shrink. On a grid of half orders from 2 to 12 (the tool prints the 21 values), every third difference of the maps' δ is positive and each is smaller than the one before. The table's values, on the grid of whole orders where it has them, give third differences that change sign at six of seven steps (−0.27, +0.13, +0.10, −0.18, +0.29, −0.34, +0.41, −0.44); its odd orders seem to sit on a curve of their own, as Vieru noticed.

How we think the difference arose

δ is the largest eigenvalue in size of the linearised doubling operator, which takes a small change h of the fixed function g to

$$(L\,h)(x) = \alpha\,[\,h(g(x/\alpha)) + g'(g(x/\alpha))\,h(x/\alpha)\,].$$

In words: the change made at the point g(x/α), plus the change made at x/α carried through the slope of g, all multiplied by α. Because α is negative, for $x > 0$ the point x/α lies on the hump's left side. The change h is written on the basis $|x|^{dj}$, $j = 0, 1, 2, \ldots$; at x/α each basis term is $|x/\alpha|^{dj}$. Written instead in signed powers, $(x/\alpha)^{dj}$, a term is the same for even d and changes sign for odd d whenever j is odd. The fixed-point equation that gives α also evaluates g at x/α, but there g is written in $|x|^{dj}$, as the reconstruction keeps it, so α is untouched; Briggs's printed α agrees with ours to 7×10^{-37} at order 3. Built with signed powers, our operator gives the table's δ to 2×10^{-29} at order 3, to all 15 decimal places we compared at orders 5 and 7, and at orders 9 and 11 to 4×10^{-13} and 3×10^{-11}, the limit of 36 coefficients there.

Could the printed number still be a true eigenvalue, answering some other question? The true operator can act on changes that are the same on both sides of the hump (even in x) and on changes that are opposite (odd in x); at order 3 it has no eigenvalue at 5.968 in either kind, and the nearest, 0.117 away, is δ itself.

Who came before

Feigenbaum found the constants and showed δ to be the largest eigenvalue of the linearised doubling operator (Feigenbaum 1978, 1979; cited for the history, not read here); Coullet and Tresser found the same universality independently (Tresser and Coullet 1978); Lanford (1982) proved the conjectures for the quadratic maximum, d = 2, and Epstein (1986) gave proofs that the fixed point exists beyond the quadratic case (both cited, not read here). Briggs computed the table for orders 2 to 12 to high precision (Briggs 1991, 1997); van der Weele, Capel and Kluiving (1987) studied α and δ as functions of the order (abstract read; full text not read). Briggs's Table 2.2 lists, under Christiansen, Cvitanović and Rugh (1990), their long value at order 2 and, in the same box, rougher values at orders 2 to 10; their letter (read through a web reader) gives δ only at order 2, so the rougher values seem to come from a source whose label the table lost. McCartney (2020) computed δ from superstable orbits to period 2^{32} for integer and non-integer orders from √2 to 10, found values that differ from Briggs's at every odd order, tested his own calculation four ways for numerical error, and concluded that the discrepancy more likely lay in the earlier work; his paper names no cause (read in its accepted manuscript). Moon's Wolfram Demonstration (2013) found its odd-order values slightly different from Briggs's and asked for an explanation. Vieru (2008) asked whether anyone had noticed that the odd orders' values lie on a different smooth curve from the even orders'. Finch's errata to *Mathematical Constants*, in its issue of 2015 and in its latest (arXiv:2001.00578 v4, 2024, read through a web reader), leaves the order-3 entry as printed.

Limits

The maps' values rest on our own programs and on programs written separately by a fresh run of the same AI assistant: independent code, not a second team. We have read Briggs's thesis table, not his code. Our literature search was limited to free records. The values here are for the maps $f(x) = 1 - \mu|x|^d$ exactly as written; we make no claim about any other family. The corrected values are McCartney's before they are ours; what this note adds is a repeat of his reading with our own code, a second route, the corrected value at order 11, and a proposed cause.

The tool

Every number this note computes is printed by FEIGENBAUM_ODD_ORDERS.py, which ships with it (Python 3 with the mpmath library; it ran in under eight minutes here): the equation's α and δ at orders 2 to 12 at 32 and 36 coefficients, with the signed-power values at odd orders (Table 1); order 3 in full (Table 2); the superstable parameters at orders 3 and 5 (Table 2); the true operator's even and odd parts at order 3; δ on the half-order grid and the third differences of both curves; δ at the orders e and π; and the maps' superstable gap ratios depth by depth at the orders where Delbourgo and Kenny gave values. The separate helper's programs and their outputs ship beside it, unchanged. Numbers quoted from the literature (Briggs's, McCartney's, Delbourgo and Kenny's) are copied, not computed, and are marked so where they appear.

The mirror maze

A stranger and his AI ran through a mirror maze with a map they had made of its mirrors. Almost every mirror showed the shape they expected, but not all of them. At first they wondered whether their own map was wrong, or whether their own shape had changed, so they checked the map again, and had a fresh copy of the AI draw it again from nothing. Then they noticed that the false mirrors were all false in the same way, a way no one, as far as they could find, had explained. As they taped a note to the exit door for the next visitor, with a copy of their map and their explanation of the false mirrors, they found a small notice already there: another visitor had seen that some mirrors were false and had said so, though not why, and the plan by the maze's entrance had never been changed. They left their note beside it anyway, hoping it would help the next person enjoy the maze more.

A sign.

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Declaration

This is a preprint without peer review, and its search of the literature is limited: sources read only in part, or not at all, are marked so above. It was built with Claude (Anthropic), which developed and ran the tool, kept the records and drafted the text, as the disclosure below says.

Tool and computational resource disclosure

Tools used. Claude (Anthropic), an AI model, took part as a research partner: it wrote and ran the tool, kept the records and drafted the text. A separate, fresh run of the same model, given the problem with no number and none of our code, wrote the independent programs described above. All numerical results were produced by tools written in Python with the mpmath library, which ship with this note. Typesetting used pandoc and wkhtmltopdf.

Responsibility. Responsibility for the correctness and adequacy of the arguments and results presented here, and for the completeness and accuracy of the citations, rests entirely with the named human author. No automated system is an author of this work. Claude is named on the title line because that is how the work was made; naming a partner is not making it an author. This follows the Leiden Declaration on Artificial Intelligence and Mathematics (2026), which asks that responsibility stay with the human authors and that the use of automated tools be disclosed; this section is that disclosure.

Verification performed. In the working record behind this note, which is not deposited with it and so cannot be checked from here, every prediction was written down, with the bar it had to clear, and fixed by checksum before the run that judged it. The maps' values were read by two methods that share no code, and again by the independent programs. The note was built and read back against its source, its pages looked at, and read whole by a fresh reader given no target and no limit, whose findings were each checked against the record before anything changed.

Attribution limits. The corrected values are McCartney's (2020) before they are ours. The cause offered here, one sign, is claimed only as not found stated in a limited review of free records; we have not seen the program that made the published table.

Computational resources. A cloud workspace on ordinary hardware; the tool ran in under eight minutes there.